

API TEST CASES

Test Case ID: API-001

Title: Verify flight search API returns available flights for valid origin, destination, date, and passenger count

Feature: Flight Search API

Endpoint: GET /api/v1/flights/search

Preconditions:

- The flight service API is up and running.
- Flight inventory exists in the database for route NYC to LAX on 2024-03-15.
- Valid authorization/API key (if required by environment) is included in headers.

Steps:

1. Send a HTTPS GET request to /api/v1/flights/search with the query parameters:
 - from: NYC
 - to: LAX
 - date: 2024-03-15
 - passengers: 2
2. Validate the HTTPS response status code.
3. Validate the response body structure and field data types.
4. Verify the total number of flight objects returned in the response array.

Expected Result:

HTTP Status Code: 200 OK

Response Format: application/json

Response Body Structure:

- Returns a JSON array containing flight objects.
- Each flight object includes:
 - id (string / integer)
 - price (number / object)
 - duration (string / integer)
 - airline (string / object)

Sample Response:

JSON

```
[
  {
    "id": "FL-101",
    "airline": "SkyWays",
    "price": 299.99,
    "duration": "5h 30m"
  },
  {
    "id": "FL-102",
    "airline": "AeroJet",
    "price": 310.00,
    "duration": "5h 45m"
  }
]
```

Execution Details:

- **Actual Result:** The request returned HTTPS 200 OK with a JSON array containing 8 valid flight objects, matching all required schema fields (id, price, duration, airline).
- **Status:** PASS

Test Title: Verify successful creation of a flight booking with valid payload

Feature: Booking Management API

Preconditions:

- API service is running and accessible.
- The flight (flightId: "FL123") is active, available, and open for booking.
- The passenger (passengerId: "P456") exists in the system database.
- Authorization headers or authentication tokens (if required) are valid.

Request Parameters:

- **Endpoint:** POST /api/v1/bookings
- **Method:** POST
- **Headers:**
 - Content-Type: application/json
 - Accept: application/json
- **Request Body:**

JSON

```
{  
  "flightId": "FL123",  
  "passengerId": "P456",  
  "paymentMethod": "credit_card"  
}
```

Steps:

1. Send a POST request to /api/v1/bookings with the valid JSON request body.
 2. Verify that the HTTPS response status code is 201 Created.
 3. Validate that the response headers contain Content-Type: application/json.
 4. Check the JSON response payload to confirm bookingId is generated and returned as a non-empty string.
 5. Verify that the status field in the response body equals "pending".
 6. Validate that additional payload parameters match the requested flight and passenger details.
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Expected Results

- **HTTP Status Code:** 201 Created
- **Response Body Structure:**
 - Contains a valid bookingId (string).
 - Contains status with value "pending".

Sample Response:

JSON

```
{  
  "bookingId": "BTX789",  
  "flightId": "FL123",  
  "passengerId": "P456",  
  "status": "pending",  
  "createdAt": "2026-09-14T17:29:34Z"  
}
```

Execution Details

- **Actual Result:** The request returned HTTPS 201 Created with payload {"bookingId": "BTX789", "status": "pending", ...}. The bookingId BTX789 was successfully generated.
- **Status:** PASS

Test Title: Verify flight search API handles invalid request parameters correctly

Feature: Flight Search API - Error Handling

Preconditions:

- The API service is running and accessible in the test environment.

Request Parameters:

- **Endpoint:** GET /api/v1/flights/search
 - **Method:** GET
 - **Query Parameters:** ?from=&to=LAX&date=invalid-date (*missing origin code and malformed date string*)
 - **Headers:** Accept: application/json
-

Steps

1. Send a GET request to /api/v1/flights/search using the invalid parameters: ?from=&to=LAX&date=invalid-date.
 2. Inspect the HTTP status code returned by the server.
 3. Verify that the server rejects the request with an HTTPS 400 Bad Request status code.
 4. Validate that the response body contains a clear error payload.
 5. Verify that the error message in the payload reads "Invalid parameters" (or contains a specific field validation error message).
-

Expected Results:

- **HTTPS Status Code:** 400 Bad Request
- **Response Payload:**

JSON

```
{
  "error": "Bad Request",
  "message": "Invalid parameters",
  "status": 400
}
```

Execution Details:

- **Actual Result:** The request returned HTTPS 200 OK with an empty array [] instead of returning a 400 Bad Request error response.
- **Status:** FAIL
- **Defect Log:** Reported bug — API fails to validate query parameters (from is empty, date format is invalid) and gracefully error out.

Test Title: Verify authenticated user can successfully retrieve their profile details

Module / Feature: User Management API / Authentication

Preconditions:

- API service is running and accessible.
- A valid user account exists in the database.
- A valid Bearer authentication token (JSON Web Token or session token) is generated for the user.

Request Parameters:

- **Endpoint:** GET /api/v1/user/profile
- **Method:** GET
- **Headers:**
 - Authorization: Bearer <valid_auth_token>
 - Accept: application/json

Steps:

1. Send a GET request to /api/v1/user/profile including the valid Authorization token in the request header.
2. Verify that the server responds with HTTP status code 200 OK.
3. Validate that the response header specifies Content-Type: application/json.
4. Inspect the returned JSON payload to confirm it contains the authenticated user's details.
5. Verify that mandatory profile fields—email, name, and preferences—are present and populated with correct data types.

Expected Results:

- **HTTPS Status Code:** 200 OK
- **Response Body Structure:**
 - Contains a user object with user details.
 - Required attributes included: email (string), name (string), and preferences (object/array).

Sample Response:

JSON

```
{
  "id": "USR-9876",
  "name": "Juana Rodriguez",
  "email": "juana.rodriguez@example.com",
  "preferences": {
    "currency": "USD",
    "seatPreference": "window",
    "mealPreference": "vegetarian"
  },
  "createdAt": "2025-01-15T08:30:00Z"
}
```

Execution Details:

- **Actual Result:** The request returned HTTPS 200 OK with a complete user profile object containing email, name, preferences, and unique user identifier.
- **Status:** PASS